

POISSON MODEL FOR PAIRED COMPARISONS APPLIED TO SOCCER TOURNAMENTS

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Summary: The majority of recognised paired comparison models (such as the Bradley-Terry model, the Thurstone-Mosteller model, and the regression models and their variations) assume that the underlying distribution is continuous. In sporting events the scoring system is, however, by nature discrete. This fact implies that continuous models for statistical analysis of data obtained from these games or tournaments are inadequate for accurate analysis. The following article is a unique attempt to apply well-motivated discrete distributions in order to estimate the strengths of soccer teams. By utilising a mathematical model we are also able to justify these observations.

1. Introduction

The method of paired comparisons is a statistical technique normally applied when the characteristic of interest is immeasurable. Therefore, a method of comparing items by means of subjectivity rather than measurements is required. Utilising the method of paired comparisons, one item in the given set is compared to another item in the given set. It is then indicated which of the two items is preferred with respect to the characteristic of interest. When sufficient pairs are compared, the results are processed to provide ranking of

the items according to a certain method. The most popular models of paired comparisons include the Bradley-Terry model, the Thurstone-Mosteller model, the row sum method, ANOVA-based methods, etc. (David, 1988). Examples of practical applications of the method are ranking tennis players (e.g. for seeding purposes), food taste testing, colour comparisons, effectiveness of new/generic drugs, ranking sports teams in tournaments, social preferences and many more.

Mosteller (1951a, b, c) has suggested the following approach to designing probabilistic models for paired comparisons. Each object under study is associated with a random variable X_i . The random event $\{X_i < X_j\}$ is interpreted as follows: $\{X_i < X_j\} \equiv$ "single comparison of object i with object j resulted in favour of object j ". In that which follows we shall refer to the distributions of the X_i 's as underlying distributions. In particular, the Thurstone-Mosteller model assumes that

$$X_i \sim N(a_i, \sigma^2),$$

i.e. underlying distributions are normal distributions with common variances. It was shown (Stern, 1990) that another important model, the Bradley-Terry model (as well as many others), also fits into the frame of Mosteller's approach with independent gamma distributions with different scale parameters as underlying distributions. Thus all existing paired comparison models (except the row sum method that is distribution-free by nature) have continuous underlying distributions. The major disadvantage of continuous models with respect to sporting events is that they cannot accommodate the concept of a "draw", as the probability of a tie, $\{X_i = X_j\}$ is evidently zero. Attempts to accommodate the possibility of a tie not only eliminate the indisputable convenience of the Thurstone-Mosteller approach, but also lead to additional

complications to the model such as the necessity to introduce and estimate so-called tie-parameters (Glenn & David, 1960).

The first Bayesian approach to paired comparisons was developed by Davidson and Solomon (1973). The two statistical models considered by them, are the multibinomial and the Bradley-Terry models. For the multibinomial model, preference probabilities are suggested. When these probabilities are unconstrained, a problem arises when defining a ranking of the objects based on the preference probabilities. Once suitable constraints are placed on the preference probabilities, the search for the Bayes rule corresponding to a specific loss function poses even greater problems, thus ranking was only dealt with in terms of the Bradley-Terry model. In the Bradley-Terry model, Davidson and Solomon suggested that, if the experimental design and prior distribution satisfy the criterion of posterior balance, then for a large class of acceptable loss functions, a Bayes ranking of the worth parameters is explicitly calculated.

Leonard (1977) introduces a method to complement the Davidson and Solomon (1973) method. Davidson and Solomon's Bradley-Terry model does not readily allow incorporation of prior knowledge of the likelihood of parameters that are related in an autoregressive fashion or alternatively depend on some explanatory variables. To incorporate these options, Leonard introduces methods based upon logistic transformations and employs multivariate normal distributions for the transformed parameters. This new method has additional advantages and is more flexible in its applications.

Chen and Smith (1984) also consider the Bradley-Terry model and obtain closed form expressions for the worth and treatment effect parameters. The technique used by Chen and Smith is marginalization, which utilises only the

relevant part of the data together with the prior information to determine an expression for each estimator.

Chebotarev (1988a, b, c) proposes a Bayesian model of paired comparisons leading to the generalised row sum method for incomplete paired comparison outcomes. The generalised row sum method has an attractive combination of properties, notably simplicity, unrestricted data structures, and unique outcome. This technique involves ranking in terms of descending order of row sums

$$s_i = \sum_{j=1}^n x_{ij},$$

where x_{ij} (result of comparing items i and j) might be expressed as some difference between items i and j , or as values $-1, 0, 1$, depending on which of the two is preferred. The row sum method does not assume any underlying distribution for the data. This is an advantage in many practical instances, but has the evident disadvantage of providing only ranking (no statistical forecasting, hypothesis testing, etc.).

Rejuvenating interest in the field of paired comparisons is evident in recent Bayesian Analysis publications. An example of such a publication is Gelman, Carlin, Stern & Rubin (1995), where the authors suggest Bayesian analysis of the multinomial paired comparison data by fitting data to a Poisson regression model (p. 394). Their findings are illustrated by means of results of chess tournaments.

2. Introduction to our new model

When considering the many possible applications of paired comparisons, we concluded that there exists a need for discrete distributions in model construction for paired comparisons. Our approach is the first which may be

successfully applied to any discrete distribution, specified prior or appropriately suspected prior, in paired comparison models. We believe that it is due to the utilisation of the Bayesian approach that our models may be set up for any underlying continuous or discrete distribution.

In this paper, we promote a new method, which aims to develop a mathematical model to explain the observations, and not merely to rank the items. This leads to more scope and flexibility. We deny the ranking as a primal goal because of the following reasons. Ranking can be obtained

1. by the values of the parameter(s) and
2. by considering the two probabilities, namely $P(\text{Item A preferred to B})$ and $P(\text{Item B preferred to A})$.

If case (1) is selected for ranking purposes and more than one parameter is appropriate for the characteristic according to which ranking may be obtained, the question arises as to which we shall select as the criterion for ranking. If case (2) is selected for ranking purposes, which appears more natural, confusion about the characteristic for ranking would not take place as the mathematical model to be developed takes all parameters into account. However, the second case may result in so-called circular triads (i.e. when A is preferred to B, B is preferred to C and C is preferred to A). That is, ranking will no longer be "linear" ranking. Therefore, we deny the ranking as a prime objective and replace it with the objective of identifying the model (underlying distribution) which best explains available observations.

Once the mathematical model has been obtained, we provide a check for the adequacy of the fit of the model to the data and the plausibility of the model for the purpose for which the model will be utilised. This checking of the model is

a vital part of the statistical analysis as Bayesian prior-to-posterior inferences assume the whole structure of the probability model and can thus yield false inferences if the model is inadequate or invalid (Gelman, Carlin, Stern & Rubin, 1995).

To conclude, the model we introduce is based on the following principles:

1. n objects are given, each of them associated with a random variable, X_i ($i = 1, 2, \dots, n$).
2. The distributions of the X_i 's, the underlying distributions, are completely unknown or only known to a certain extent.
3. The sample (observations) are functions of the pairs of the random variables, e.g. $\text{sign}(X_i - X_j)$, $X_i - X_j$, $\frac{X_i}{X_j}$, etc.
4. The objective is to find such underlying distributions from the given class of distributions that best fit the observations.

3. Poisson model for paired comparisons

In the field of probability theory, the Poisson distribution is known to describe the number of occurrences of a certain event during a specified period of time (Feller, 1966). In soccer, a game between two teams is played within a certain specified time period and the points scored for each team recorded. Thus it would be appropriate to consider the points scored by each team, for a particular game, to be a Poisson distribution, and thus the difference in scores per game to be the difference between two Poisson distributions. From this, we can conclude that the use of a discrete underlying distribution, in particular the Poisson distribution, is well motivated. Also note that, in soccer, a tie in score is

possible. This can be suitably accommodated within the Poisson distributions. The other paired comparison models, for example the Bradley-Terry model, Thurstone-Mosteller model and regression models, seem inadequate as they have been designed essentially for continuous data. Our attempt is basically the first attempt to fit a discrete model to paired comparison data.

We observe our pairwise comparisons as the difference between two Poisson distributions, i.e.

$$X_i \sim \text{Poisson}(\lambda_i), \quad i = 1, 2, \dots, n$$

and

$$X_j \sim \text{Poisson}(\lambda_j), \quad j = 1, 2, \dots, n$$

and we observe

$$K = X_i - X_j.$$

The random variable K has the following probability function:

$$f_{K|\lambda_i, \lambda_j}(k|\lambda_i, \lambda_j) = e^{-\lambda_i - \lambda_j} \left(\frac{\lambda_i}{\lambda_j} \right)^{\frac{k}{2}} I_{|k|} \left(2\lambda_i^{\frac{1}{2}} \lambda_j^{\frac{1}{2}} \right), \quad k = 0, \pm 1, \pm 2, \pm 3,$$

where $I_k(\cdot)$ is the modified Bessel function (Feller, 1966).

In conventional Bayesian theory we use a Gamma prior for estimation of a Poisson distribution parameter. Therefore, we follow this convention and assume as prior information that

$$\lambda_i \sim \text{Gamma}(\alpha_i, \beta_i)$$

and

$$\lambda_j \sim \text{Gamma}(\alpha_j, \beta_j)$$

where

$$\lambda_i \text{ and } \lambda_j \text{ are independent.}$$

Independence of these two parameters is a reasonable assumption and common practice in other paired comparison models (Stern, 1990). Thus

$$G(\lambda_i, \lambda_j) = \frac{\beta_i^{\alpha_i} \beta_j^{\alpha_j} \lambda_i^{\alpha_i-1} \lambda_j^{\alpha_j-1} e^{-\beta_i \lambda_i} e^{-\beta_j \lambda_j}}{\Gamma(\alpha_i) \Gamma(\alpha_j)} \quad \lambda_i, \lambda_j > 0$$

The posterior distribution of λ_i, λ_j is obtained using the following:

$$f_{\lambda_i, \lambda_j|k}(\lambda_i, \lambda_j|k) = \frac{G(\lambda_i, \lambda_j) \cdot f_{k|\lambda_i, \lambda_j}(k|\lambda_i, \lambda_j)}{\int_0^\infty \int_0^\infty G(\lambda_i, \lambda_j) \cdot f_{k|\lambda_i, \lambda_j}(k|\lambda_i, \lambda_j) d\lambda_j d\lambda_i}$$

Due to the symmetry of the above distribution, it is possible to limit the consideration to the case where k is positive. If a negative difference in score is observed, the technique to apply is to switch the players and take the absolute value of the observed difference, and proceed with k positive.

The posterior distribution of λ_i, λ_j is:

$$f_{\lambda_i, \lambda_j|k}(\lambda_i, \lambda_j|k) = c \frac{\lambda_i^{\alpha_i-1+\frac{k}{2}} \lambda_j^{\alpha_j-1-\frac{k}{2}} I_k\left(2\lambda_i^{\frac{1}{2}} \lambda_j^{\frac{1}{2}}\right)}{e^{\lambda_i(1+\beta_i)+\lambda_j(1+\beta_j)}}$$

where

$$c = \frac{(1+\beta_i)^{\alpha_i+k} (1+\beta_j)^{\alpha_j} \Gamma(1+k)}{\Gamma(\alpha_j) \Gamma(\alpha_i+k) {}_2F_1\left[\alpha_j, \alpha_i+k, 1+k, (1+\beta_i)^{-1} (1+\beta_j)^{-1}\right]}$$

As we can see, the prior distribution is not conjugate. After finding the posterior distribution, Bayes estimators were obtained for λ_i and λ_j , using the squared error loss function:

$$\hat{\lambda}_i = E[\lambda_i|k]$$

$$= \frac{(\alpha_i+k) {}_2F_1\left[\alpha_j, 1+\alpha_i+k, 1+k, (1+\beta_i)^{-1} (1+\beta_j)^{-1}\right]}{(1+\beta_i) {}_2F_1\left[\alpha_j, \alpha_i+k, 1+k, (1+\beta_i)^{-1} (1+\beta_j)^{-1}\right]}$$

and

$$\hat{\lambda}_j = E[\lambda_j|k]$$

$$= \frac{\alpha_j {}_2F_1 \left[1 + \alpha_j, \alpha_i + k, 1 + k, (1 + \beta_i)^{-1} (1 + \beta_j)^{-1} \right]}{(1 + \beta_j) {}_2F_1 \left[\alpha_j, \alpha_i + k, 1 + k, (1 + \beta_i)^{-1} (1 + \beta_j)^{-1} \right]}$$

where ${}_2F_1(\cdot)$ is the Generalised Hypergeometric Function. The squared error loss function is minimised and thus to determine Bayes estimators for the parameters λ_i and λ_j , namely $\hat{\lambda}_i$ and $\hat{\lambda}_j$, $i, j = 1, 2, \dots, n$, the first moments of the joint posterior distribution are required. Thereafter, to determine Bayes estimators $\hat{\alpha}_i$, $\hat{\alpha}_j$, $\hat{\beta}_i$ and $\hat{\beta}_j$, $i, j = 1, 2, \dots, n$ the first and the second moments of the joint posterior distribution are utilised as follows :

$$\frac{\alpha_i}{\beta_i} = E[\lambda_i] = \mu_{1i} \quad \text{and} \quad \frac{\alpha_j}{\beta_j} = E[\lambda_j] = \mu_{1j}$$

$$\frac{\alpha_i(\alpha_i + 1)}{\beta_i^2} = E[\lambda_i^2] = \mu_{2i} \quad \text{and} \quad \frac{\alpha_j(\alpha_j + 1)}{\beta_j^2} = E[\lambda_j^2] = \mu_{2j}.$$

Solving these equations, the following is obtained:

$$\hat{\alpha}_i = \frac{\mu_{1i}^2}{\mu_{2i} - \mu_{1i}^2} \quad \hat{\beta}_i = \frac{\mu_{1i}}{\mu_{2i} - \mu_{1i}^2}$$

$$\hat{\alpha}_j = \frac{\mu_{1j}^2}{\mu_{2j} - \mu_{1j}^2} \quad \hat{\beta}_j = \frac{\mu_{1j}}{\mu_{2j} - \mu_{1j}^2}$$

These new estimators $\hat{\alpha}_i$, $\hat{\alpha}_j$, $\hat{\beta}_i$, $\hat{\beta}_j$, $\hat{\lambda}_i$ and $\hat{\lambda}_j$ are then stored and processed when required as priors in future pairs. This process is repeated for all the pairs of observations in the experiment and we finally obtain the best estimators for $\hat{\lambda}_i$ and $\hat{\lambda}_j$, $i, j = 1, 2, \dots, n$, the estimated strengths of the teams.

Conventional methods of paired comparisons (e.g. Bradley-Terry method, row sum method) require balanced data, that is, the number of games between

every pair of contestants should be the same. Such tournament designs are known as round-robin tournaments, of which the most common is one where each pair meets twice, namely once at the home field of each team. In some other classical methods (e.g. Thurstone-Mosteller method), the requirement of balanced data could be eliminated, while the number of observations for each pair should be relatively large. However, in big international tournaments, neither of the two requirements (balanced data, sufficient observations for each team) can be met. In fact, in such tournaments some pairs of teams do not even meet once.

The use of the Bayesian technique in our approach compensates for the non-balanced data by means of using relatively accurate prior estimates known from previous analyses of similar data. E.g., for soccer we know that estimates for the λ_i 's usually lie in the interval $(0, 2.5)$. The Bayesian approach also secures acceptable accuracy of the estimates even for relatively small sets of data. In the test runs of our algorithms for real data and simulated data, we obtained precise results even for relatively poor accuracy of priors if there were 2–4 observations per team (not per pair!).

Another feature of our approach is that the more recent comparisons have more weight than earlier ones. This is essential in sport as the strength of a team does change in time and the more recent achievements should carry more weight.

EXAMPLE

We applied the above model to the 1995 African Cup of Nations soccer results.

Estimates for λ -parameters (strength parameters) for the Poisson distributions are given in the following statistical ranking, together with the ranking according to official final tournament ranking:

Teams	Statistical ranking	Tournament ranking
Zambia	: 2.46300	South Africa
South Africa	: 1.78711	Tunisia
Zaire	: 1.53305	Zambia
Ghana	: 1.41602	Ghana
Tunisia	: 1.38018	Algeria
Gabon	: 1.36544	Gabon
Algeria	: 1.01687	Egypt
Egypt	: 0.97872	Zaire
Liberia	: 0.70116	Liberia
Ivory Coast	: 0.60238	Cameroon
Cameroon	: 0.53710	Angola
Sierra Leone	: 0.40236	Ivory Coast
Mozambique	: 0.00041	Mozambique
Angola	: 0.00012	Sierra Leone
Burkina Faso	: 0.00008	Burkina Faso

To prove that previously made assumptions (independence of teams' strengths, independence of match outcomes, Poisson underlying distribution, order of elements in pairs not essential, etc.) are valid for soccer data, we will consider the hypothesis

$$H_0 : X_i \sim \text{Poisson}(\hat{\lambda}_i) \quad i = 1, 2, \dots, 15$$

where $\hat{\lambda}_i$'s are given in the above table.

The goodness-of-fit was tested by means of a simple modification of the Chi-square test, using calculated expected frequencies according to the model and comparing these to the observed frequencies. The computed test statistic value of 8.897 was obtained and a critical value for $\alpha = 0.05$ is $\chi^2 = 9.49$.

Since $8.897 < 9.49$, we accept the null hypothesis and conclude, at the 5% level of significance, that the presented model does adequately describe the sample behaviour of the observations, i.e. describes the soccer results.

It is important to note that the list of estimates is not merely rankings, but mathematical estimates of the parameters. Thus forecasts can be made for future games between two specified teams that did not meet in this tournament, using the mathematical estimates of the strength parameters.

To illustrate, consider South Africa and Zambia:

South Africa's estimated strength, $\lambda_2 = 1.78711$.

Zambia's estimated strength, $\lambda_1 = 2.463$.

Assume that X_2 represents the score obtained by South Africa and X_1 the score obtained by Zambia. Then

$$\begin{aligned} & P \text{ (South Africa wins a future game against Zambia)} \\ &= P(X_2 > X_1) \\ &= 0.279483 \end{aligned}$$

$$\begin{aligned} & P \text{ (Zambia wins a future game against South Africa)} \\ &= P(X_2 < X_1) \\ &= 0.529454 \end{aligned}$$

$$\begin{aligned} & P \text{ (South Africa and Zambia tie in a future game)} \\ &= P(X_2 = X_1) \\ &= 0.191062 \end{aligned}$$

When considering $P(X_2 - X_1 = k)$, for $k = 0, \pm 1, \pm 2, \pm 3, \dots$, the forecast of the most likely difference in score, in Zambia's favour, is 1. The probability of such a difference, $P(X_2 - X_1 = -1)$, is 0.19529.

Some interesting remarks on the statistical ranking obtained:

- * The first four teams in the tournament ranking appear in the first five statistical rankings.
- * Zambia achieved the highest estimated strength, but did not reach the final. Zambia had two wins, each by a large point difference of 4 (Burkina Faso and Sierra Leone). Zambia lost only once, to Tunisia (relatively strong team) in the semi-final by two points and were thus knocked out of the tournament.
- * South Africa won the tournament, but did not receive the highest statistical ranking. Except for the win against Cameroon (weak team), South Africa did not win by a large difference in score, thus their ranking did not increase dramatically at any stage. Throughout the tournament South Africa had home advantage, especially in the final in Johannesburg where crowd support was overwhelming. On the day of the final South Africa had a "nothing to lose, all to gain" attitude, which encouraged them to play at their limit for the spectators.
- * Tunisia played in the final, but only achieved an estimated strength of 1.38018 (fifth largest ranking). Tunisia lost in the final against South Africa (strong team), by two points and as it was mentioned before, the latest observations have the largest weight and this loss pushed Tunisia down in ranking.

4. Conclusion

From the above example we can conclude that the Poisson underlying distribution is suitable for soccer results. The scoring system in games such as soccer, polo and hockey, are all similar in the fact that one point is scored for each occurrence of a goal score. The model suggested in the paper is specifically good for soccer-type data. Our attempts to use the model for games with different scoring systems (e.g. rugby) have shown poor fit.

5. References

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